Forecasting with hmmTMB: An Example Using Old Faithful Data

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2026-08-28

1 Introduction

This vignette demonstrates using hmmTMB for forecasting with a hidden Markov model (HMM). The Old Faithful dataset, included in base R, contains observations of eruption durations and waiting times between eruptions. We model the eruption durations as a two state Markov process, influenced by waiting times through non-linear effects in the transition probabilities.

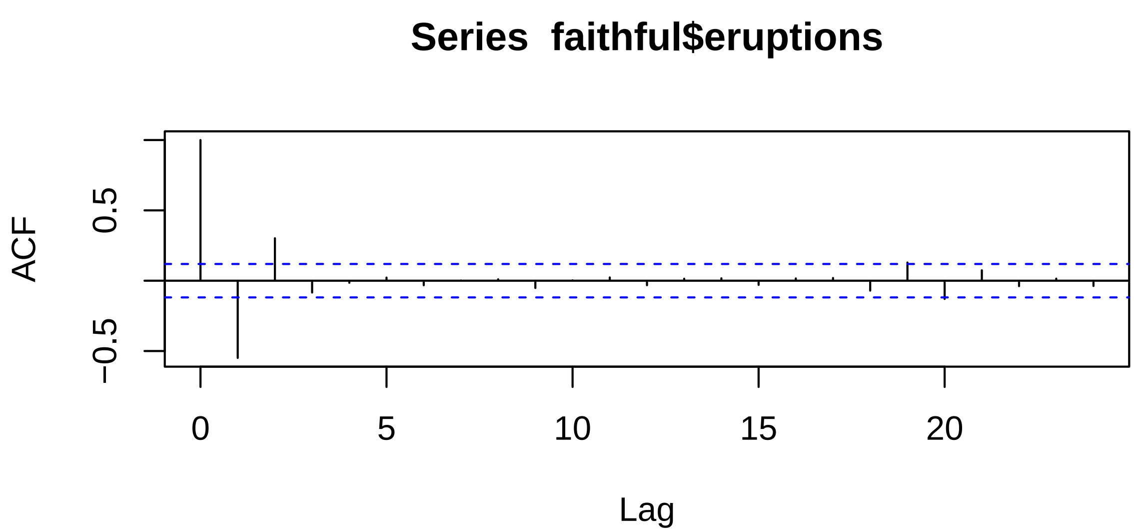
1.1 Data Exploration

First, load the necessary library and explore the data.

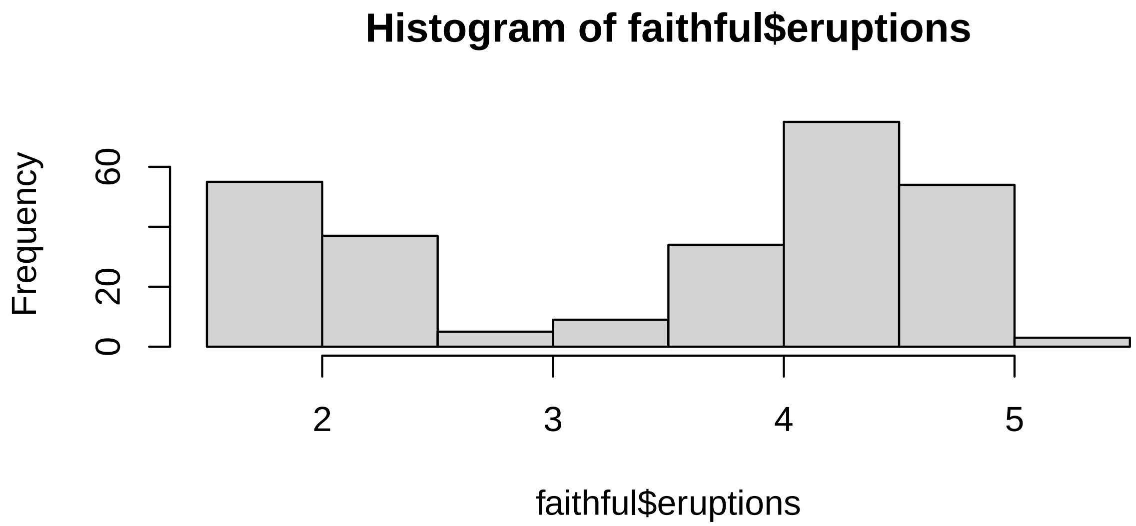
```
library(devtools)
library(ggplot2)
library(ggridges)
load_all("../../hmmTMB")
data("faithful")
```

We examine the autocorrelation, distribution, and relationship between variables.

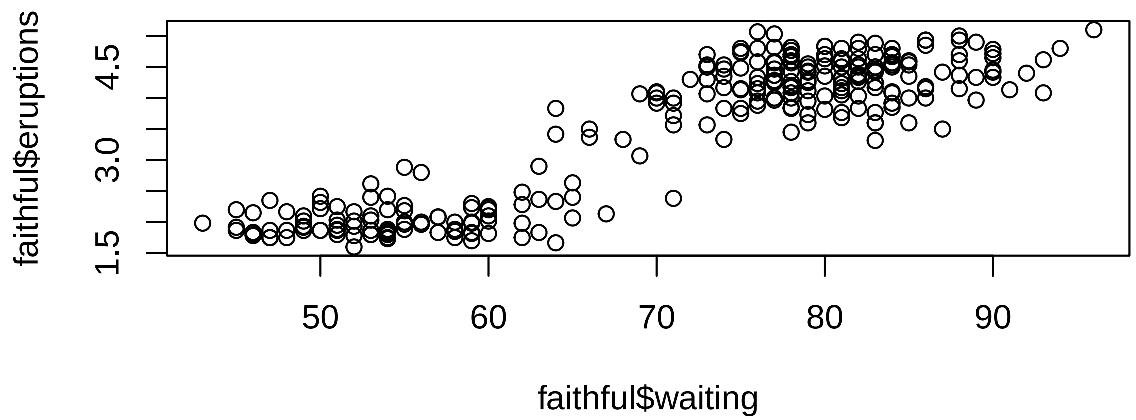
```
acf(faithful$eruptions)
```



```r
hist(faithful$eruptions)
```




```r
plot(faithful$waiting, faithful$eruptions)
```



The autocorrelation plot shows a temporal pattern to eruption durations, where duration has a -0.5 correlation with lag -1 and +0.3 correlation with lag -2, likely caused by large eruptions depleting the water supply and shortening subsequent eruptions. The histogram indicates a bimodal distribution, suggesting two distinct eruption types. The scatter plot reveals a positive relationship between waiting time and eruption duration, indicating that waiting time is a useful predictor of eruption duration.

1.2 Data Preparation

Split the data into training and forecasting sets, using 80% for training.

```r
training_fraction <- 0.8
n_training <- floor(nrow(faithful) * training_fraction)
training_df <- data.frame(
  ID = rep(1, n_training),
  waiting = faithful$waiting[1:n_training],
  eruptions = faithful$eruptions[1:n_training]
```

```
)
forecast_df <- data.frame(
  ID = rep(1, nrow(faithful) - n_training),
  waiting = faithful$waiting[(n_training + 1):nrow(faithful)],
  eruptions = faithful$eruptions[(n_training + 1):nrow(faithful)]
)
```

1.3 Model Specification

Define a 2-state Markov chain with non-linear effects of waiting time on transitions using smoothing splines, and normal distributions for eruption durations.

```
hid_model <- MarkovChain$new(
  data = training_df,
  n_states = 2,
  formula = ~ s(waiting, k = 10, bs = "cs")
)
obs_model <- Observation$new(
  data = training_df,
  n_states = 2,
  dists = list(eruptions = "norm"),
  par = list(eruptions = list(
    mean = c(0, 0),
    sd = c(1, 1)
  ))
)
# Update to suggested initial parameters
obs_model$update_par(par = obs_model$suggest_initial())
hmm <- HMM$new(
  hid = hid_model,
  obs = obs_model
)
hmm$fit(silent = TRUE)
```

2 Forecasting

Create a forecast object using the fitted model and forecast data, starting from the last state distribution of the training period.

```
forecast <- Forecast$new(
  hmm = hmm,
  forecast_data = forecast_df,
  starting_state_distribution = "last"
)
```

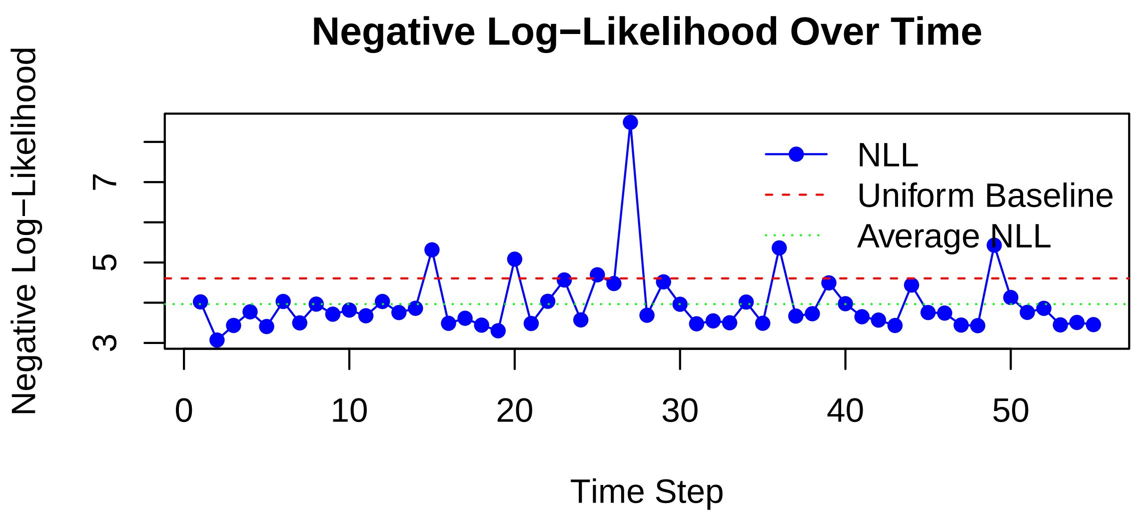
2.1 Evaluation with Negative Log-Likelihood

Compute NLL for the forecasts and plot the results.

```r
# Define the observation variable
observation <- "eruptions"

# Extract parameters
eval_range <- forecast$eval_range()[[observation]]
forecast_dist_matrix <- forecast$forecast_dists()[[observation]]
step_size <- eval_range[2] - eval_range[1]
time_steps <- seq(ncol(forecast_dist_matrix))

# Find indices of closest evaluation points to true values
true_value_indices <- sapply(
  forecast_df[[observation]],
  function(true_val) which.min(abs(eval_range - true_val))
)

# Extract PDF values
pdf_values <- forecast_dist_matrix[cbind(true_value_indices, time_steps)]

# Compute NLL (approximating probability mass)
nll <- -log(pdf_values * step_size)

# Uniform baseline NLL
total_range <- max(eval_range) - min(eval_range)
nll_baseline <- -log(1 / length(eval_range))  # Alternative: -log(step_size / total_range)

# Average NLL
average_nll <- mean(nll, na.rm = TRUE)
```

Plot the NLL over time steps.

```r
plot(
  nll,
  type = "o",
  col = "blue",
  pch = 16,
  xlab = "Time Step",
  ylab = "Negative Log-Likelihood",
  main = "Negative Log-Likelihood Over Time"
)
abline(h = nll_baseline, col = "red", lty = 2)
abline(h = average_nll, col = "green", lty = 3)
legend(
  "topright",
  legend = c("NLL", "Uniform Baseline", "Average NLL"),
  col = c("blue", "red", "green"),
  lty = c(1, 2, 3),
  pch = c(16, NA, NA),
  bty = "n"
)
```

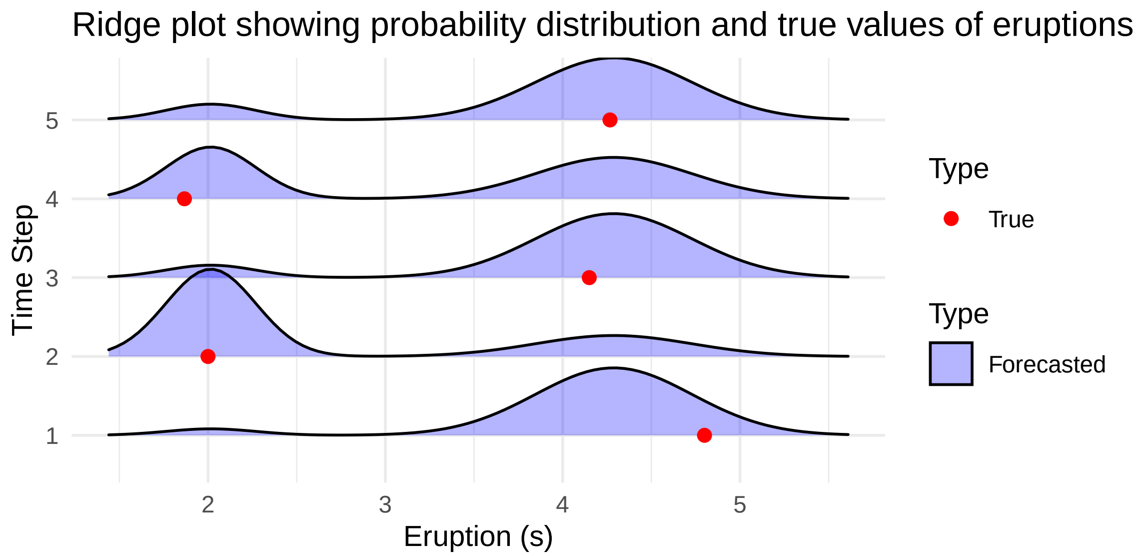


The negative log-likelihood (NLL) plot shows the model's performance over time steps. The blue line represents the NLL for each time step, while the red dashed line indicates the performance of a uniform predictor with no predictive power, and the green dashed line shows the average NLL across all time steps. The plot shows a strong consistent performance even at long time horizons. The data includes several outliers with poor predictive scores.

3 Visualizing Forecast Distributions with Ridge Plots

Use ggplot2 and ggridges to visualize forecast distributions and true values.

```
# Timesteps ahead to be plotted
forecast_timesteps <- c(1, 2, 3, 4, 5)

eval_range <- forecast$eval_range()[[observation]]
ridge_data <- data.frame(
  x = rep(eval_range, length(forecast_timesteps)),
  time = factor(rep(forecast_timesteps, each = length(eval_range)), levels = forecast_timesteps),
  density = c(forecast$forecast_dists()[[observation]][, forecast_timesteps])
)
true_df <- data.frame(
  time = factor(forecast_timesteps, levels = forecast_timesteps),
  x = forecast_df[[observation]][forecast_timesteps]
)

ggplot() +
  geom_ridgeline(data = ridge_data,
                 aes(x = x, y = time, height = density, fill = "Forecasted"),
                 alpha = 0.3) +
  geom_point(data = true_df,
             aes(x = x, y = time, colour = "True"),
             size = 2, inherit.aes = FALSE) +
  scale_fill_manual(name = "Type", values = c("Forecasted" = "blue")) +
  scale_colour_manual(name = "Type", values = c("True" = "red")) +
  labs(title = paste("Ridge plot showing probability distribution and true values of", observation), x
  theme_minimal() +
  guides(fill = guide_legend(override.aes = list(alpha = 0.3)))
```



This ridge plot displays the forecasted probability densities at selected time steps, overlaid with the true eruption durations.